Efficiency and acceleration in the accumulation of children's early vocabulary

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Abstract

Children learn hundreds of words over the first few years of their life. This process begins slowly but increases until only a few exposures to a word are necessary for a child to learn it. Yet there are also wide variations between children, and the distribution across children has not not been accurately characterized. Here we present a statistical model describing this process as accelerating accumulation, with children varying in both the slope and intercept of their accumulation. Using data from three languages, this model provides a good characterization of longitudinal growth and provides insights into the sources of age and socioeconomic variation. Using dense observations of children's early language environment, the model also can be used to estimate how much variation between children is due to differences in language input, finding estimates that converge with recent meta-analyses. In contrast to children, large language models are not well-described by this model: they show standard accumulator dynamics and very limited variability across a number of dimensions.

Children's early vocabulary grows very rapidly during early childhood, with large variation between children (**fenson1994?**; **fenson2007?**; Frank et al. 2021). The growth has theoretical, practical, and methodological importance. From a theoretical perspective, growth mechanisms shed light on children's fundamental learning mechanisms and how they change across childhood (McMurray 2007; **bloom2002?**; Frank et al. 2021). From a practical perspective, early vocabulary is an important precursor – and predictor – of academic achievement [(**fernald2008?**);CITES]. From a methodological perspective, vocabulary also affords a unique psychometric opportunity to study human variation because, unlike nearly all important human psychological metrics, there is a true zero point – a time before which children know no words. Thus, absolute levels of variability can be calculated.

One dominant viewpoint is that children's vocabulary growth can be characterized as a process of accumulation (McMurray 2007; **mitchell2009?**; Hidaka 2013; Mollica and Piantadosi 2017; Kachergis, Marchman, and Frank 2022). Each distinct word (type) can be conceptualized as a "bucket," and every token of language input then is a "drop" that falls its respective bucket. When each bucket is full, the word is learned (McMurray 2007; Kachergis, Marchman, and Frank 2022). These models have been used to recover interpretable differences in "bucket size," often based on word properties including concreteness and syntactic category (Hidaka 2013; Kachergis, Marchman, and Frank 2022).

The accumulator view connects with a viewpoint in which the quantity of language that children hear plays a strong causal role in the speed of their learning (Fernald and Marchman 2012). Indeed, there is systematic evidence that variation in children's input is related to variation in vocabulary [Fernald and Marchman (2012);(**weisleder2013?**);CITES]. Studies of input have identified systematic variation in *quantity* in terms of sheer amount of tokens as well as variation in *quality*. Measuring what makes input high quality has been challenging, but higher quality input likely has higher lexical diversity, longer sentences, more declarative and interrogative content relative to imperatives, and more elaborated discourse [LOTS OF CITES].

A naïve interpretation of the accumulator view would suggest that most of the variance in vocabulary growth should be related to input. Two recent meta-analyses complicate this view, however. (**anderson2021?**) showed that only a modest proportion of the variance in vocabulary outcomes is related to input quantity (XYZ) and quality (XYZ). A newer meta-analysis of quantity specifically confirms and extends the proportion of variance due to quantity to 4–7% of total variation in vocabulary (**coffey2026?**). Thus – as we might have supposed based on other research on human variation – not all variation in vocabulary between children can be attributed to input differences.

As this discussion indicates, despite the promise of the accumulator viewpoint, it has not been used successfully to understand variation between individuals. Instead, the majority of applications have focused on variation between words. Further, the rapidity of children's early word learning standards as a rebuke to this view. Sometimes a small number of examples is enough to form a lasting representation (**markson2002?**; **woodward1994?**). Perhaps only a small number of high-quality instances is sufficient for learning (**medina2010?**; **trueswell2014?**; Mollica and Piantadosi 2017), but if so, why do some words take much longer to learn?

The goal of this work is to address these puzzles through systematic investigation and extension of accumulator models. Following previous work, we consider that children's learning ability might accelerate across development (**hagihara2013?**); in the bucket metaphor, the drops get bigger as children get older. We begin by extending the psychometric accumulator model of Kachergis, Marchman, and Frank (2022): our key move is to model variation in individual children' learning efficiency and the acceleration of their learning. This extension defines a longitudinal growth model that can be fit to data, and that provides a simple reconciliation between slow accumulation and fast mapping: the number of examples of a word necessary for learning declines by multiple orders of magnitude from age one to age three.

To constrain our model, we use data from the MacArthur-Bates Communicative Development Inventories [MB-CDI; (**fenson1994?**);(**marchman2021?**)], a popular parent report instrument in which parents are asked to indicate which words their child produces and understand. Here we focus exclusively on production, which is used with older children and which tends to be a more reliable measure [Frank et al. (2021);others]. We make use of longitudinal data from Wordbank (**frank2017?**), an open repository of data from the MB-CDI.

To address questions about the relationship between language input and growth, we fit a Bayesian version of our growth model. We make use of new data resources to estimate the magnitude of linkages between input and vocabulary growth. First, we estimate plausible population values based on independent estimates of population variation in language input (**coffey2024?**). In addition, we use measures of individual children's language input from egocentric videos (**long2025?**) and day-long home recordings (**dong2025?**; **adams2018?**) to estimate the input-uptake relationship more precisely for a subsample of children.

Overall, our analysis supports a view in which children's vocabulary growth is not a pure process of accumulation but rather one that accelerates rapidly over time. We end by comparing the fit of our models on human data to their fit on large-language models. Making use of GPT-2 models trained on human developmental data, we show that – in contrast to children – these models do appear to function as pure accumulators. This analysis exposes a critical disanalogy between LLMs and human learners and suggests a potential route for understanding the vast "data gap" between children and large language models (**frank2023?**).

Model

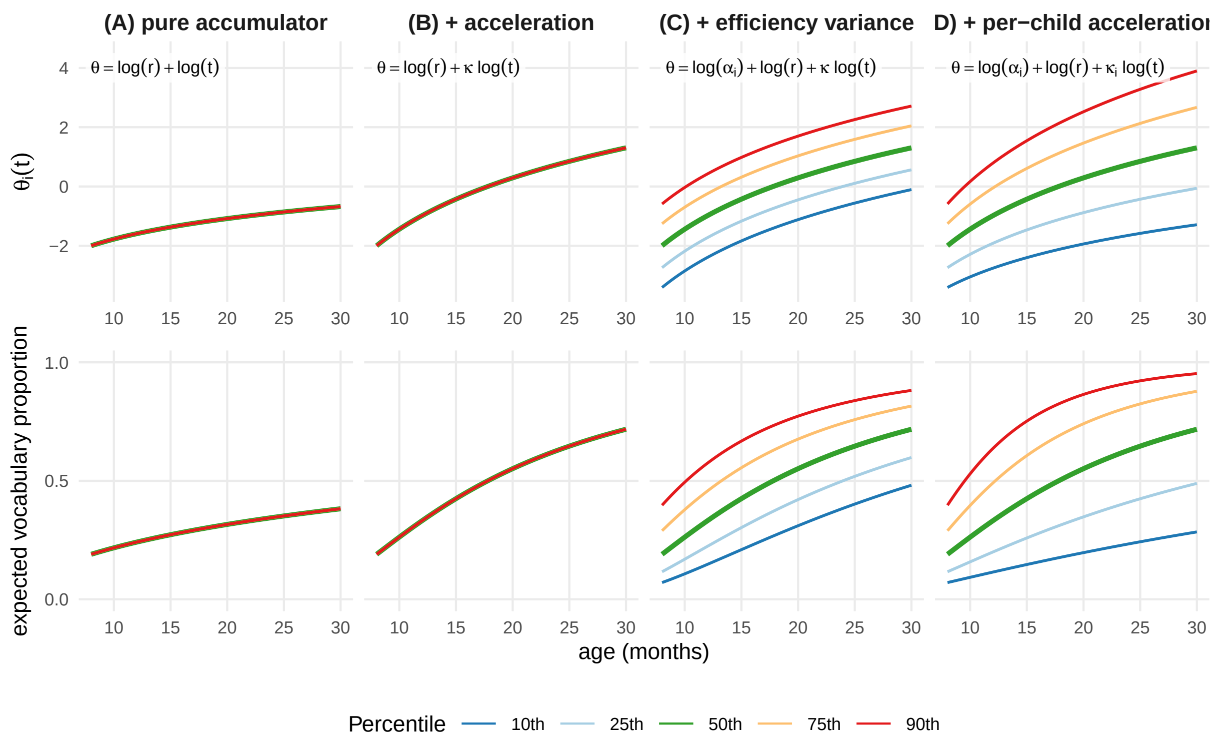

Figure 1

We present a nested set of accumulator-type models, many of which have been presented in the literature. As we show below, each refinement of the model contributes to its ability to describe the variability between children. We start with the Rasch model from psychometrics (Rasch 1960; Bond and Fox 2013), which assumes that each child i has a learning ability θ_i and each word j has a difficulty δ_j. The probability that child i produces word j is

$$P(w_{i,j} = 1 \mid \theta_i, \delta_j) = \frac{\exp(\theta_i - \delta_j)}{1 + \exp(\theta_i - \delta_j)}$$

Following Kachergis, Marchman, and Frank (2022), we can write our first model a simple Rasch-style accumulator model ("M1: pure accumulator"), specifying that θ_i is not a single parameter but instead a function of accumulation over time. In the simplest form:

$$\theta_i = \log(r_i) + \log(t)$$

where r_i is the child's rate of language input (in tokens per month) and t is the child's (age in months); note that we use log age to simulate linear accumulation because $\exp(\theta) = r \cdot t$. This form instantiates the basic hypothesis that the only thing that changes over time is more input, and the only source of variation between children is variation in rate.

We compare this model to an accelerating accumulator model, in which the amount learned from each exposure increases with time by a multiplicative factor *kappa* (which manifests as an exponent on time in the full Rasch model).

$$\theta_i = \log(r_i) + \kappa \ \log(t)$$

Neither of these models has any way to capture variation across children beyond variation in input rate. Thus, we augment the basic accelerating accumulator in two ways, rewriting it as the the sum of two individually variable quantities:

$$\theta_i = \xi_i + \zeta_i \ \log(t)$$

The intercept term $\xi_i = \log(\alpha_i) + \log(i)$ then assumes that there is both rate-related and non-rate variation in children's efficiency. Similarly, the slope term $\zeta_i = \kappa_i + \gamma_i \log r_i$ decomposes acceleration into rate-related and non-rate-related acceleration.

This final model instantiates the idea that children vary in their base rate of efficiency and their acceleration, instantiating a longitudinal growth model in which children vary on their slope and intercept. Note that, while we define the relationship of ξ_i and ζ_i to a child's input r_i, we can nevertheless fit the simplest form of the model by just allowing these parameters to vary as random child-level slopes and intercepts.

As we will show, this model provides a good description of variation in children's longitudinal growth trajectories. Figure 1 shows schematic predictions from a pure accumulator model (A), an accelerating accumulator (B) in comparison to models including child-level variation in efficiency (C) and both efficiency and acceleration (D). Each model's predictions are represented as ability scores (θ) across age and then also projected into proportion correct scores on a discrete set of vocabulary items, representing an observed accuracy score that is subject to compression at floor and ceiling.

A variety of previous work has focused on estimating the difficulty of individual words (Goodman, Dale, and Li 2008; Hidaka 2013; **roy2015?**; Braginsky et al. 2019; Kachergis, Marchman, and Frank 2022). These analyses reveal that there is predictable word-level difficulty information but that only a modest portion of the variation between words is carried by frequency information. Words can be hard for a child to acquire because they vary also in the complexity and concreteness of their meanings, their syntactic category and syntactic context, their phonology, and many other factors. In the current work, we estimate word difficulty as a free parameter, and focus instead on variation between children.

Our analysis generally depends on the availability of longitudinal data at the item level. The different models described above are not distinguishable from the cross-sectional, aggregate data that are typically considered in many analyses of early vocabulary. First, aggregate data do not allow for the estimation of individual word difficulties, thus the distribution of difficulties remains latent and different difficulty distributions can be posited to explain the accelerating shape of vocabulary growth (McMurray 2007; **mitchell2009?**). Second, cross-sectional data do not allow for accurate estimation of whether variation is due to differences in efficiency (intercept) or acceleration (slope) (see Supplemental Figure 1).

Results

Children's longitudinal growth reflects differences in both efficiency and acceleration

We first retrieved longitudinal data on monolingual, typically-developing children's vocabulary from Wordbank (**frank2017?**), an open archive of data from the MacArthur-Bates Communicative Development Inventory and its international adaptations. Table 1 shows the four longitudinal datasets in Wordbank that contained more than > 100 children with multiple observations.

Table 1: Characteristics of the datasets.

Dataset	Citation	N	Admins (mean)	Ages	Recordings
English (American)	@thal2013,@marchman2013, L. Smith	1,874	1.3	8–30 mo	—
Norwegian	@simonsen2014	1,676	1.6	8–36 mo	—
French (Quebecois)	[@CITE]	179	1.1	8–30 mo	—
Japanese	@hagihara2023	178	1.3	8–36 mo	—
BabyView	@long2025	22	5.0	6–32 mo	Headcam (n = 5,739 videos)
SEEDLingS	@dong2025	44	12.0	6–17 mo	LENA (n = 525 recordings)
AM2018	@adams2018	66	3.0	16–24 mo	LENA (n = 126 recordings)
FMW2013	@fernald2013	51	3.0	18–30 mo	LENA (n = 51 recordings)

Taking advantage of the equivalence of Rasch models to generalized linear mixed effects models (De Boeck et al. 2011), we fit the set of models shown in Figure Figure 1 as a series of logistic regressions. As described above, this initial set of fits do not attempt to estimate the relationship with input but instead simply attempt to describe the distribution of vocabulary growth curves. Each model predicted the probability of an individual child producing a particular word as a function of age (either unmodified in model A or multiplied by a constant in model B) and a random effect of word difficulty. Successive models add random intercepts (model C) and slopes (model D) by child.

Model D, which incorporated variation in acceleration and efficiency, was the best fitting model across all datasets, with AIC and BIC differences shown in Table 2. Figure 2 shows the fit of the four model variants; the increasing spread of children's trajectories with age is especially noticeable for low-vocabulary older children [who may be classified as "late talkers"; Fernald and Marchman (2012); Rescorla (2011)].[1]

[1]Following the literature, we primarily investigate linear accumulator models, which show power-law growth of ability over time because θ is a function of $\log(t)$. However, it is also possible that ability grows linearly in time, which would lead to exponential increases in the probability of word production. We fit linear-time models, but found that they were dispreferred statistically, though differences were relatively small within the restricted age range for which we have data (δAIC 169–4,138; see Supplemental Figure 2 for model fits).

Table 2

Model	Age	ΔAIC				ΔBIC			
		English	Norwegian	French (Q.)	Japanese	English	Norwegian	French (Q.)	Japanese
A. accumulator	—	1,721,195	2,299,678	66,408	95,222	1,721,143	2,299,625	66,367	95,181
B. + acceleration	linear	925,989	1,389,719	44,967	41,321	925,950	1,389,679	44,936	41,290
B. + acceleration	log	929,263	1,360,175	43,091	39,171	929,224	1,360,135	43,061	39,140
C. + efficiency var.	linear	153,291	131,294	4,746	3,864	153,265	131,267	4,726	3,843
C. + efficiency var.	log	150,361	115,789	3,407	3,239	150,335	115,763	3,386	3,218
D. + acceleration var.	linear	711	4,138	250	169	711	4,138	250	169
D. + acceleration var.	log	0	0	0	0	0	0	0	0

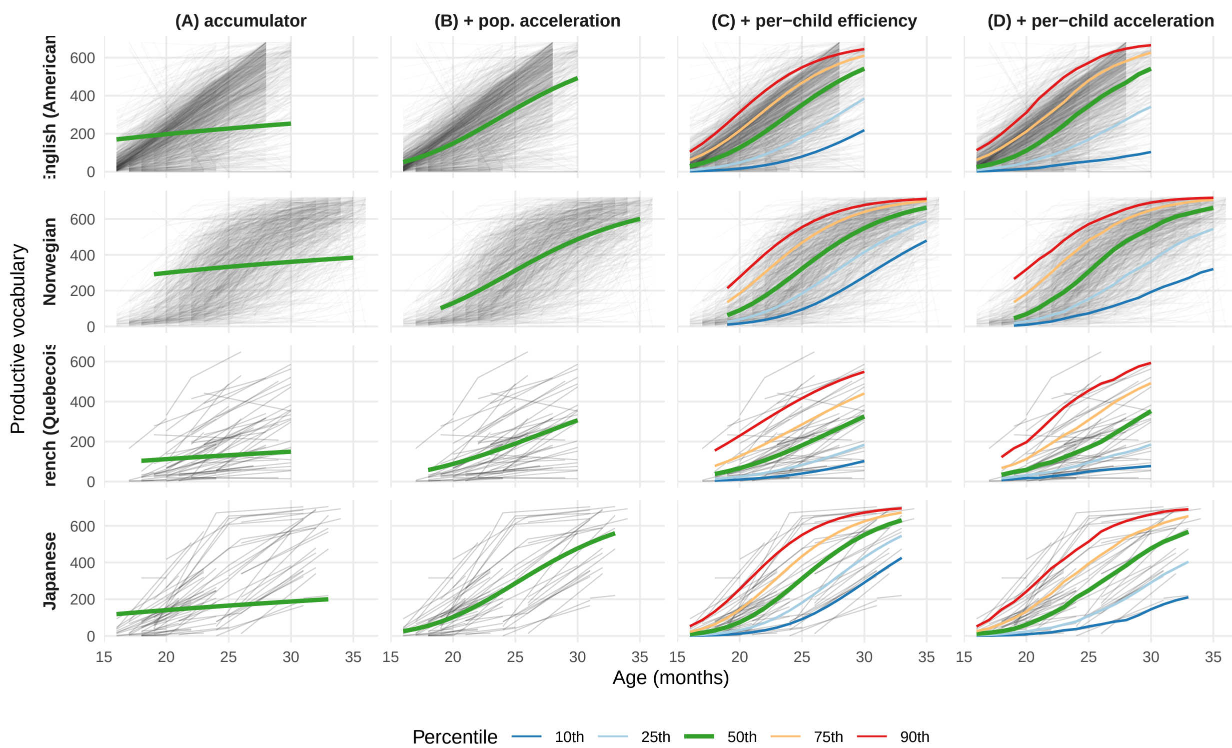

Figure 2: Individual children's longitudinal vocabulary growth trajectories across age in four languages. The same data are plotted in each column of panels, with overlaid models of increasing complexity (colored lines). Transparency is higher for English (American) and Norwegian panels to avoid overplotting.

Efficiency and acceleration relate to separate demographic predictors

Children's vocabulary is known to be related to a number of sociodemographic predictors, including sex and maternal education (which is interpreted as a proxy for socioeconomic status). In particular, female children show consistently larger vocabularies than male children across dozens of languages and countries (**erikson2012?**; Frank et al. 2021). Similarly, children of more educated mothers tend to show larger vocabularies, though this effect is somewhat more heterogeneous across countries and languages (Frank et al. 2021).

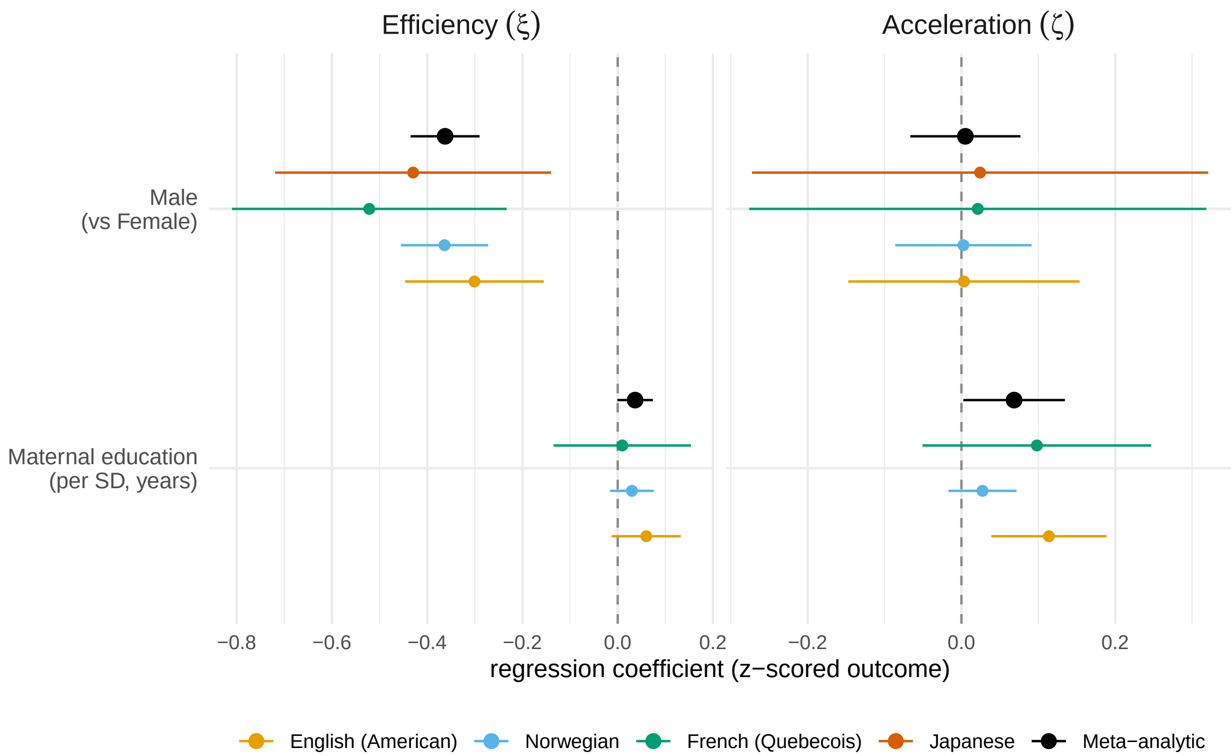

Figure 3: Regression coefficients predicting individual children's efficiency and acceleration as a function of gender and maternal education. Black dots show meta-analytic estimates from a random-effects meta analysis. Error bars show 95% confidence intervals.

Our longitudinal model offers a summary of individual children's growth in terms of both their efficiency (intercept) and their acceleration (slope). This decomposition allows us to ask whether these two demographics effects – sex and maternal education – are seen in efficiency, acceleration, or both. For each child in our analysis, we extracted the best linear unbiased prediction (BLUP) from our models. We then regressed intercept and slope terms against each demographic predictor separately for each language. Figure 3 shows the coefficients from this analysis, along with a cross-language meta-analytic estimate.

Sex effects show the predicted female advantage across all four languages, but this effect manifests itself only in efficiency ($\beta = -0.36$, $p =< .001$) not acceleration ($\beta = 0.01$, $p = .888$). In contrast, though maternal education data were only available for three languages, effects were numerically present in both measures, though stronger in acceleration ($\beta = 0.07$, $p = .042$) than in efficiency ($\beta = 0.04$, $p = .056$). One interpretation of these findings is that sex effects reflect a small constant offset in average ability. In contrast, socioeconomic differences as reflected in maternal education might continue to compound across development (Fernald, Marchman, and Weisleder 2013; Hart and Risley 2003).

Growth in ability leads to rapid changes in learning times for individual words

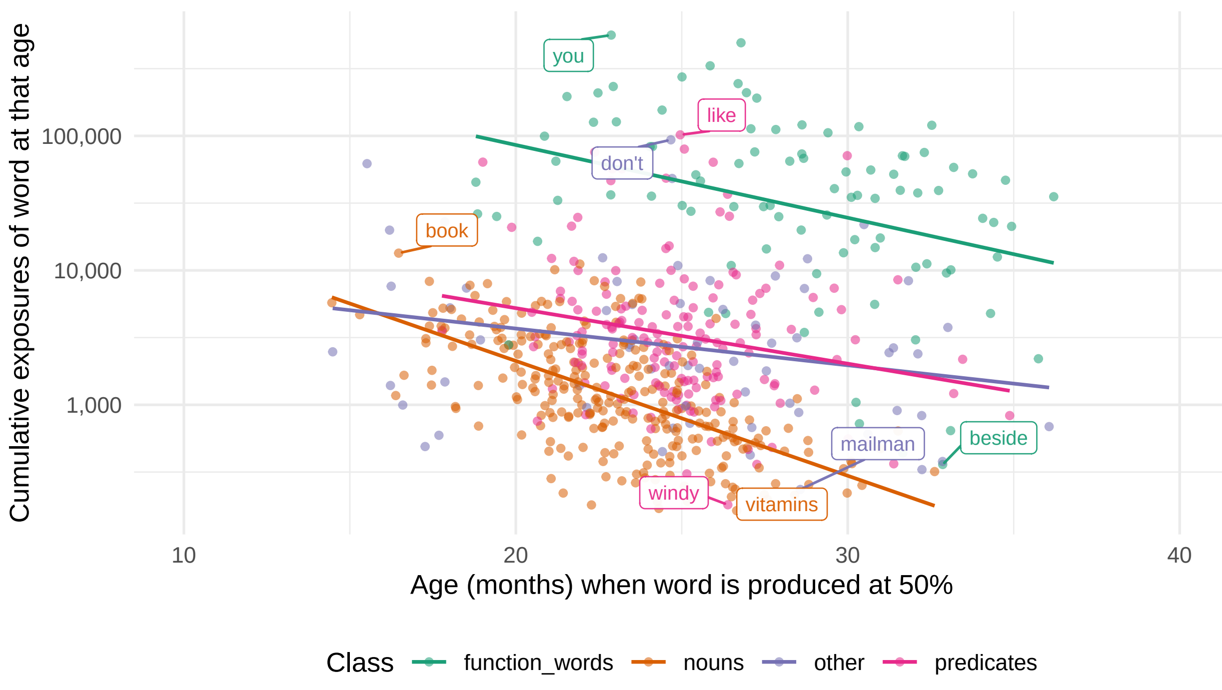

Figure 4

As an illustration of this change in efficiency, Figure 2 shows a model-derived estimate of the average number of times a child would need to hear a word before producing it (See SI: Changes in Word Efficiency), using frequency estimates from the CHILDES Corpus (**mcwhinney2020?**; **sanchez2019?**). Across different word classes (e.g., nouns, verbs, closed class words), efficiency increases exponentially, such that early-learned words such as "ball" may be heard tens of thousands of times before they are produced, while later learned words such as "vitamin" might only be heard tens of times before being produced.

Estimating input

Under a range of plausible assumptions about variation in input, all children show supra-linear scaling of growth, even accounting for accumulation dynamics.

As discussed above, one important question is how much of variance is due to variance in rate. We explore this by asking about the parameter π,

$$\pi = \frac{\sigma_\alpha^2}{\sigma_\alpha^2 + \sigma_r^2}$$

If $\pi > .5$, then the majority of variation between individuals is not due to pure accumulation rate.

In addition, 80-90% of between-child variation remains unexplained by input quantity. These conclusions are supported by analysis of longitudinal input data from BabyView (N=20) and SEEDLings (N=44) datasets, which contain child-specific estimates of input, and Peekbank (N=XYZ), which contains child-specific estimates of processing speed.

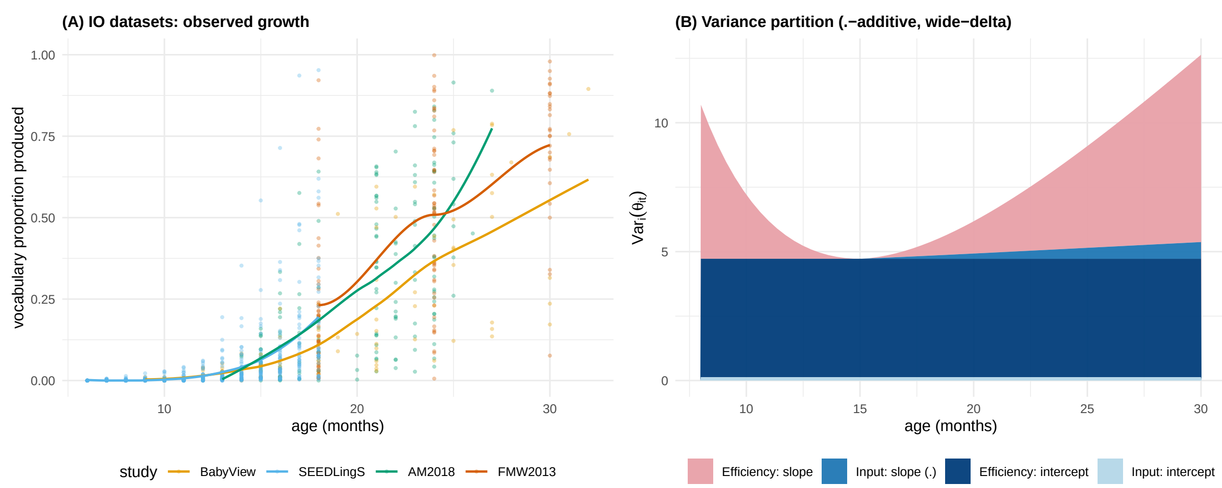

Figure 5

Thus, modeling human learning requires positing other sources of variability and acceleration. Sources of variability could include from variation in interactional input quality, genetic variation between children, environmental richness not captured by input, etc. Sources of acceleration could include increases in processing efficiency, learning-to-learn including mutual exclusivity, shape bias, syntactic bootstrapping, as well as domain general changes in perception, processing speed, memory, attention or other aspects of cognition. The key to understanding children's data efficiency may thus lie in better understanding when and how children's learning accelerates with maturation and experience.

We end by providing a quantitative comparison between within-child human learning and LLM scaling laws, showing that children's early language learning is strikingly different from LLM scaling: it shows both steeper scaling and greater variance in growth rate.

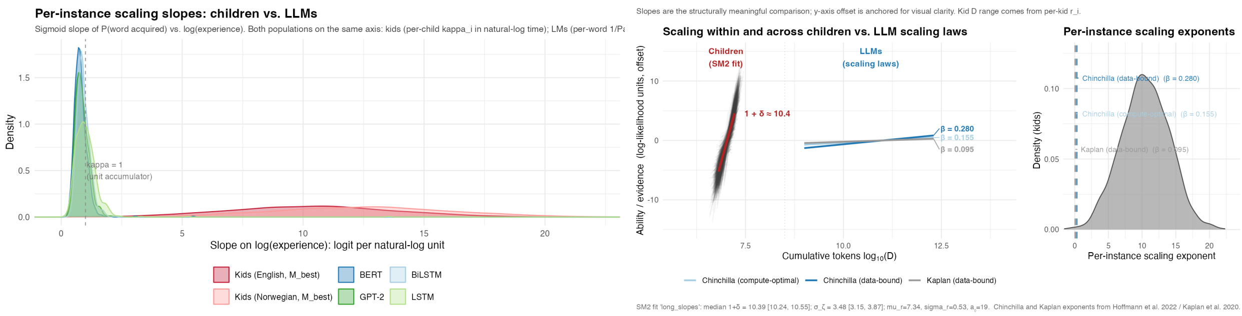

Figure 6

Discussion

What we did

Longitudinal growth models efficiency and acceleration. Two different paramters.

Where doe acceleration come from? Could come from a variety of sources, including general maturation of memory and attention, specific changes in word recognition or processing (**fernald1998?**; **frank2026?**), the increasing sophistication of inferential strategies for word learning [(**smith2002?**);(**markman1988?**);etc], or other sources.

Role of input: Coffey 2026 variance decomposition

Furthermore, how does the accumulation process relate to the rapid growth of vocabulary? On some models, accumulation by itself can produce an exponential "vocabulary explosion" while the child's own learning abilities remain constant. On other accounts, however, the child's own abilities change over development. These changes could be due to the accumulation of language allowing bootstrapping (e.g., via mutual excusivity or the shape bias), they could be through faster processing (Fernald rich get richer), or they could be via general maturation (e.g., better memory, faster processing). Or all of these.

Blingualism

International adoption

https://bergelsonlab.fas.harvard.edu/sites/g/files/omnuum3941/files/2025-01/meylan_bergelson_2021_ARL.pdf

Claude Code was used for literature review, code generation, visualization, and simulation infrastructure. All writing is original to the author.

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